```
Malawi, step 2 (2026-10-20)

 chain     seed iterations.saved acceptance.rate final.log.likelihood best.log.likelihood errored.proposals
     1 20263016             5000           0.275             -31669.6            -31651.9                 0
     2 20263017             5000           0.275             -31671.8            -31660.1                 0
     3 20263018             5000           0.275             -31686.9            -31670.7                 0
     4 20263019             5000           0.274             -31658.4            -31655.3                 0
                                    file
 mcmc_malawi_step2_chain1_2026-10-19.Rdata
 mcmc_malawi_step2_chain2_2026-10-20.Rdata
 mcmc_malawi_step2_chain3_2026-10-20.Rdata
 mcmc_malawi_step2_chain4_2026-10-20.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
        age.20.to.29.transmission.multiplier.1         age.20.to.29.transmission.multiplier.0
                                        1.984                                          1.844
                     female.to.male.multiplier                                      trate.1
                                        1.797                                          1.722
        age.30.to.39.transmission.multiplier.1         age.5.to.14.hiv.mortality.multiplier.0
                                        1.662                                          1.627
        age.0.to.14.hiv.mortality.multiplier.1        age.25.to.49.hiv.mortality.multiplier.1
                                        1.593                                          1.467
       age.25.to.49.hiv.mortality.multiplier.0        age.15.to.24.hiv.mortality.multiplier.0
                                        1.454                                          1.450
        age.40.to.49.transmission.multiplier.0                                      trate.2
                                        1.428                                          1.416
        age.15.to.19.transmission.multiplier.3                                      trate.0
                                        1.404                                          1.400
                      male.awareness.multiplier         age.20.to.29.transmission.multiplier.2
                                        1.375                                          1.321
        age.30.to.39.transmission.multiplier.0                              age.assortativity
                                        1.313                                          1.307
age.65.to.79.mortality.slope.multiplier.female                male.hiv.mortality.multiplier.1
                                        1.279                                          1.272
         age.0.to.4.hiv.mortality.multiplier.0         age.30.to.39.transmission.multiplier.2
                                        1.267                                          1.233
        age.15.to.19.transmission.multiplier.2                                      trate.3
                                        1.224                                          1.216
        age.40.to.49.transmission.multiplier.1
                                        1.213

Merged simset:
 location simulations total.weight                                          file
   Malawi        1000         1000 simset_malawi_step2_chains1234_2026-10-20.Rdata
```

Malawi step 2: likelihood

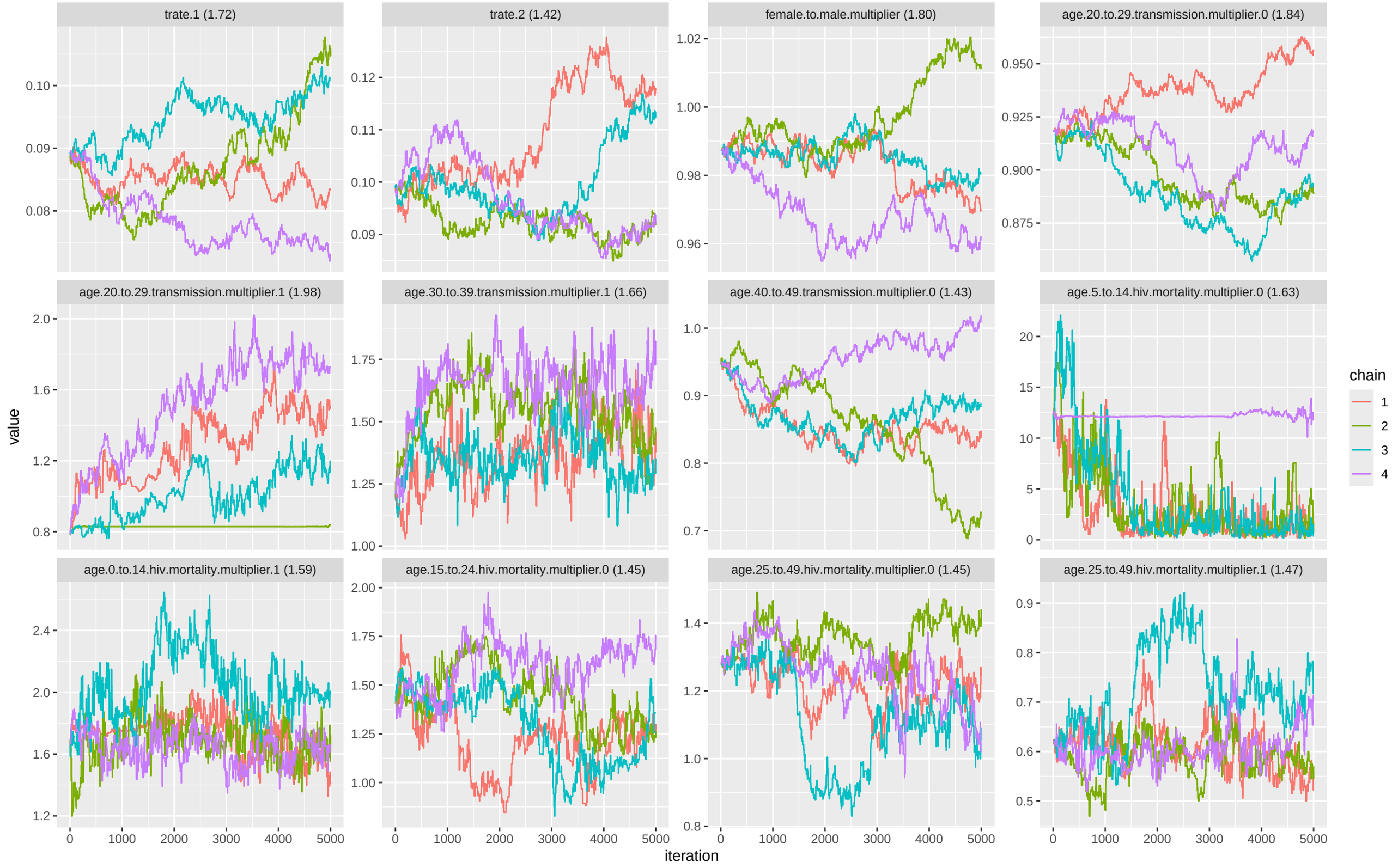
0
−10000
−20000
−30000
lue

0
25000
50000
75000
100000
iteration

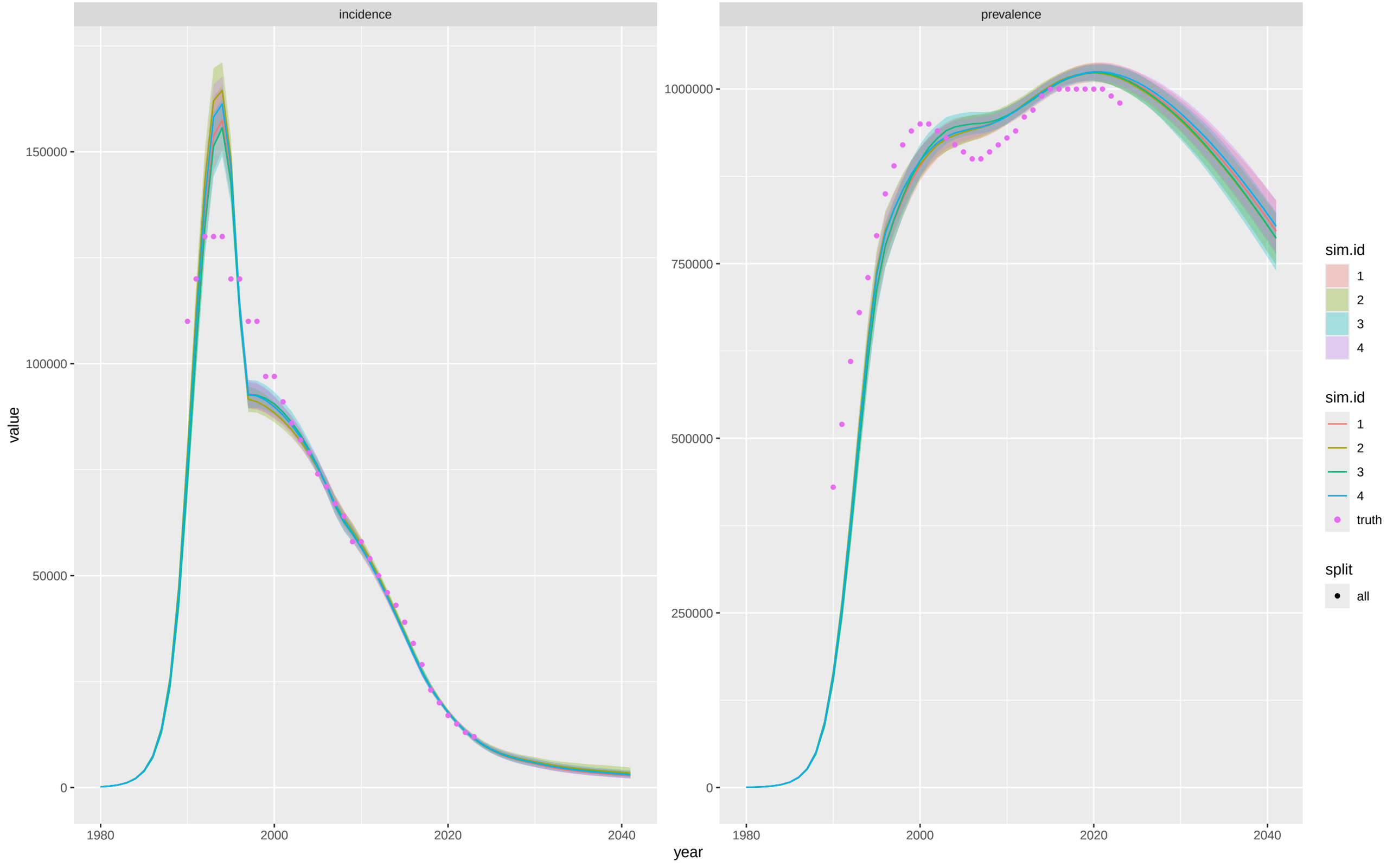
chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

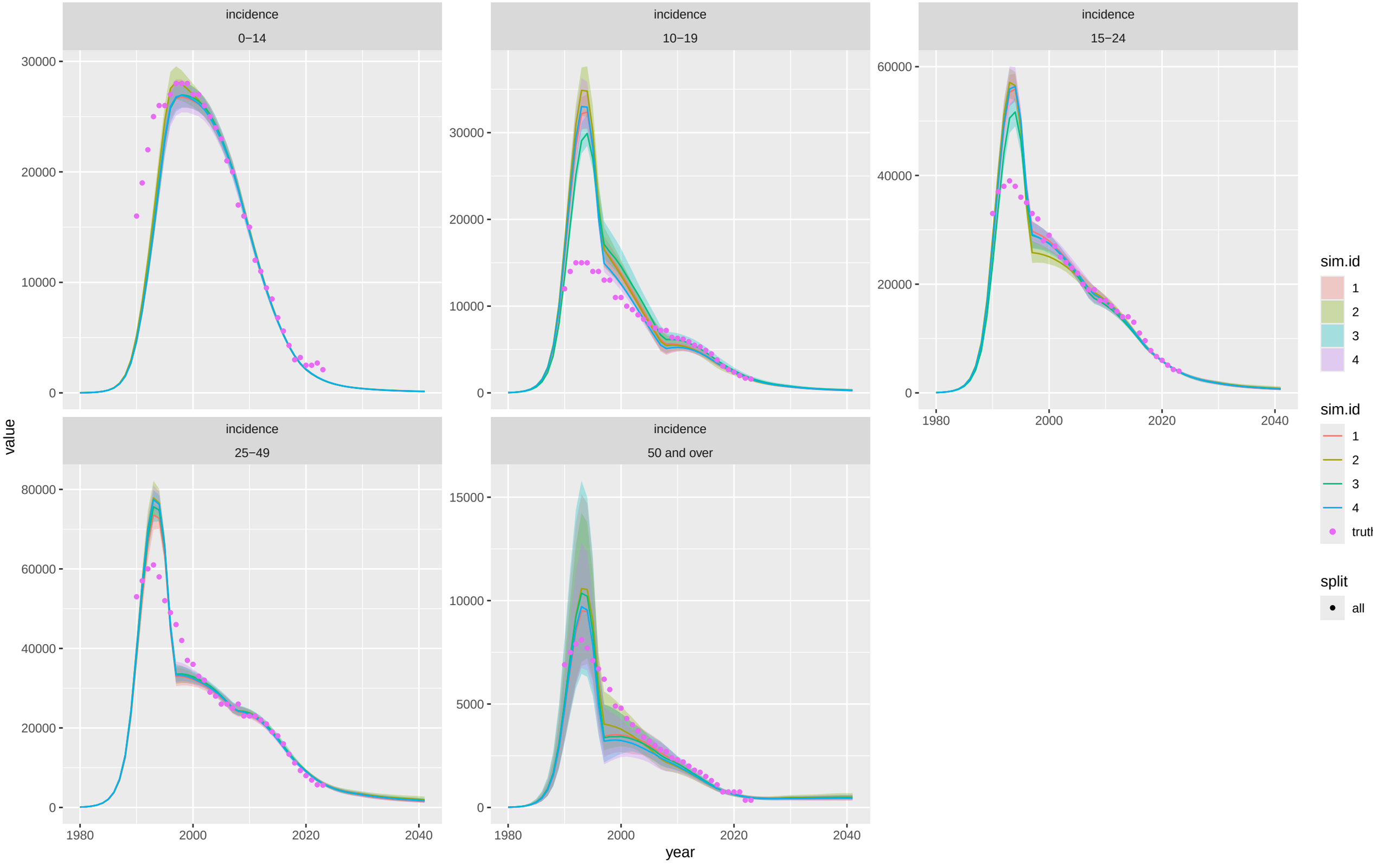
Malawi step 2: trace



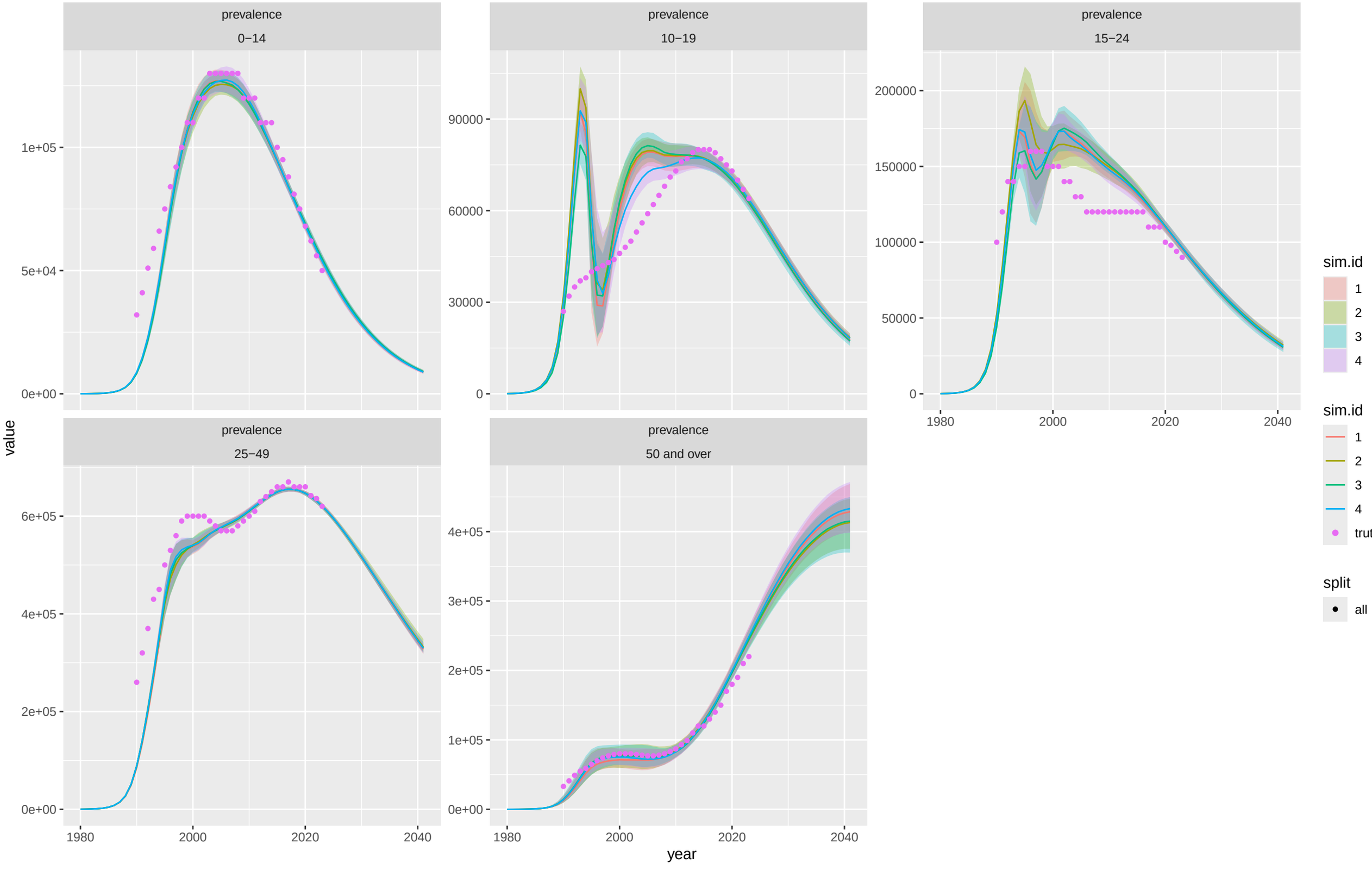
Malawi step 2: fit (one color per chain)



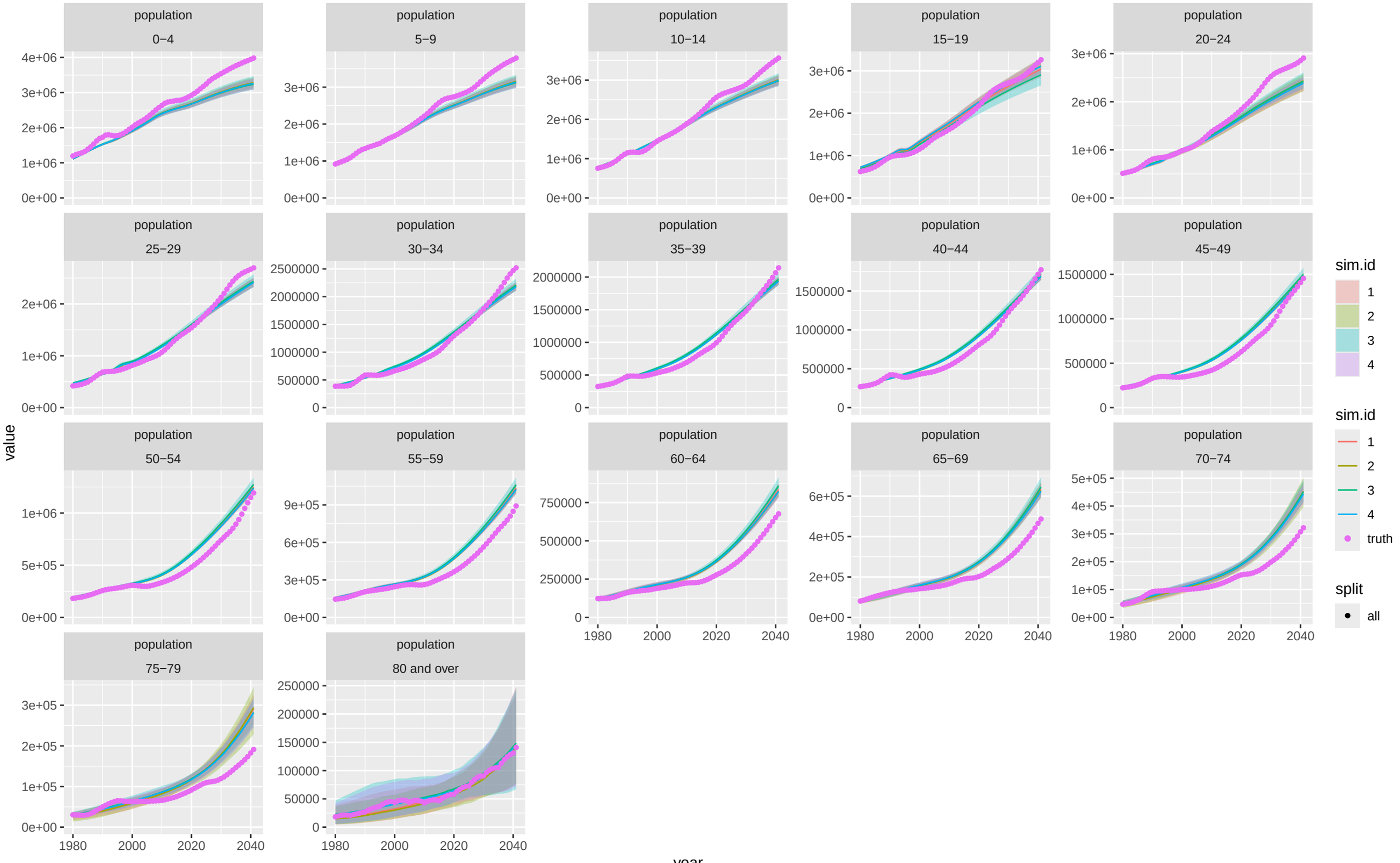
Malawi step 2: incidence.by.age (one color per chain)



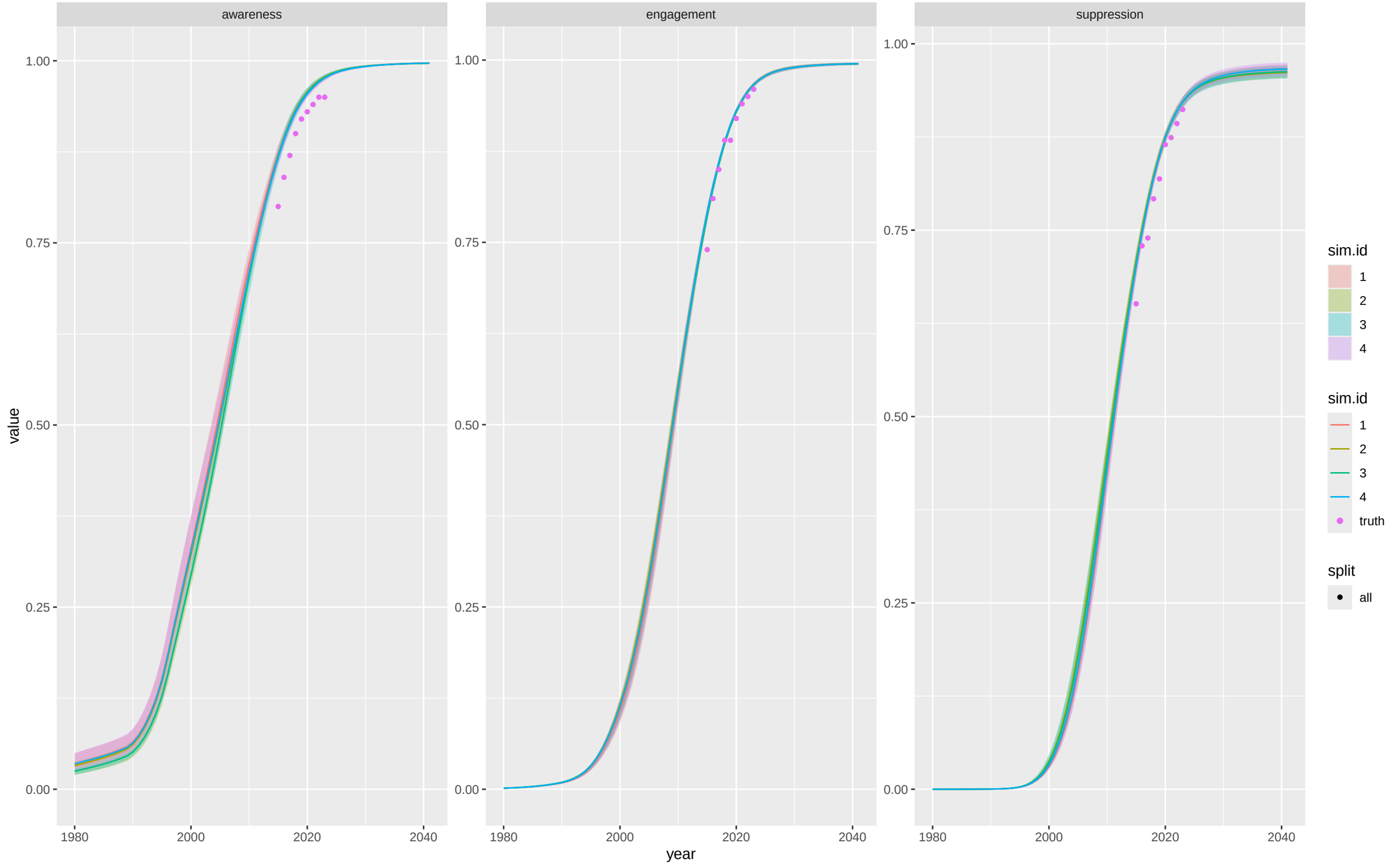
Malawi step 2: prevalence.by.age (one color per chain)



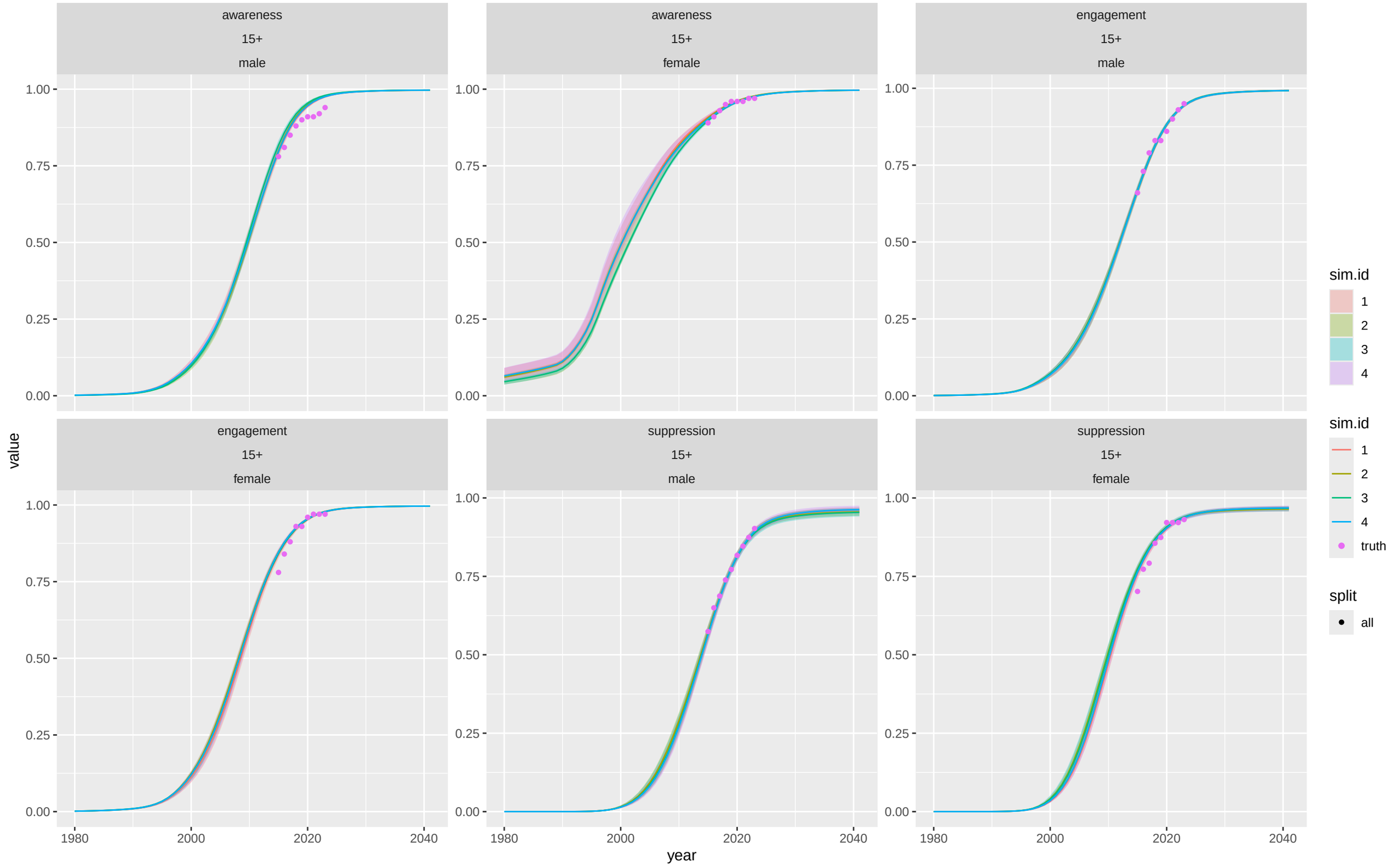
Malawi step 2: population.by.age (one color per chain)



Malawi step 2: cascade (one color per chain)



Malawi step 2: cascade.by.age.sex (one color per chain)



Malawi: fit (merged Step 2 simset)

incidence

prevalence

value

150000
100000
50000
0

1000000
750000
500000
250000
0

1980
2000
2020
2040

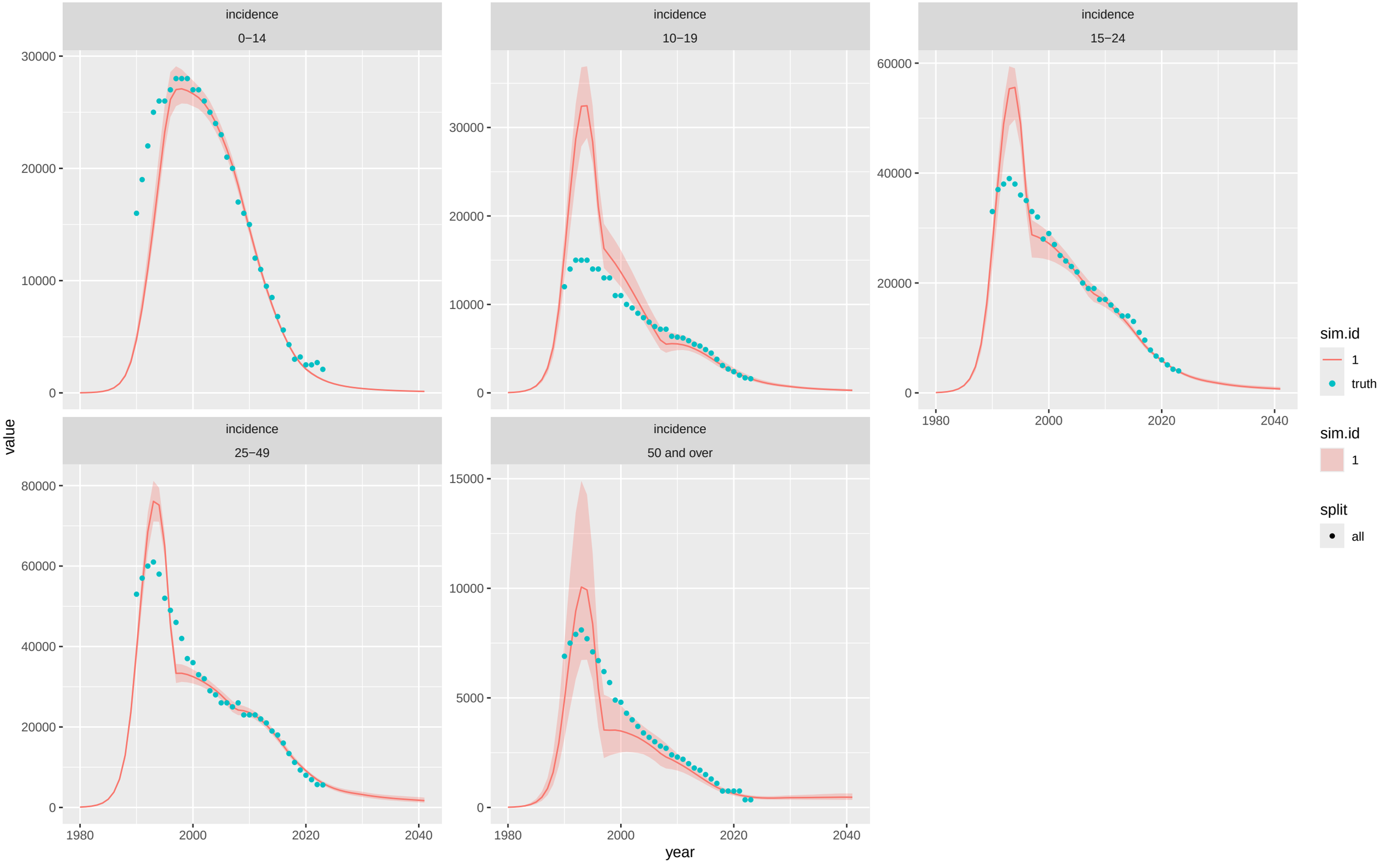
year

sim.id
1
truth

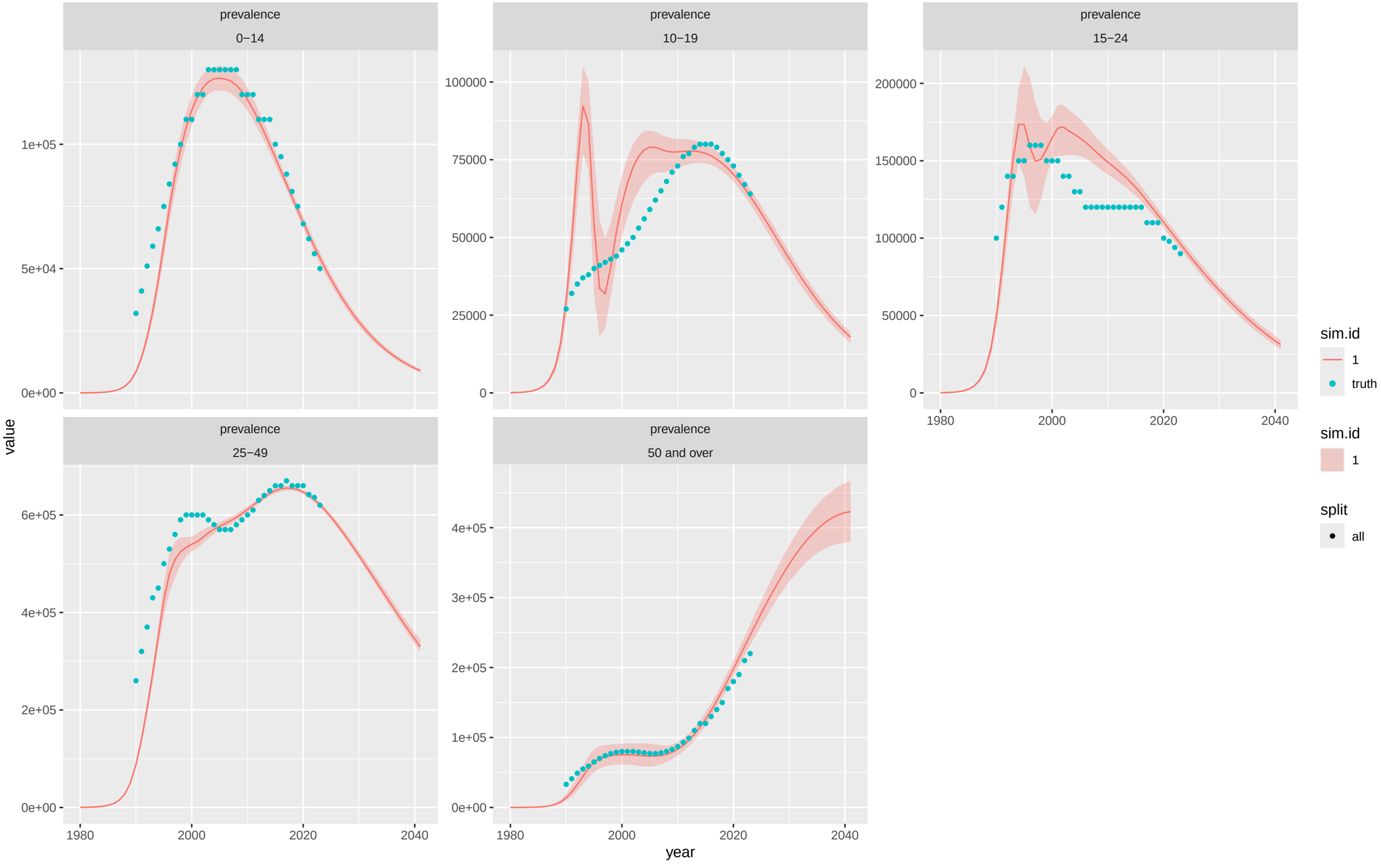
sim.id
1

split
all

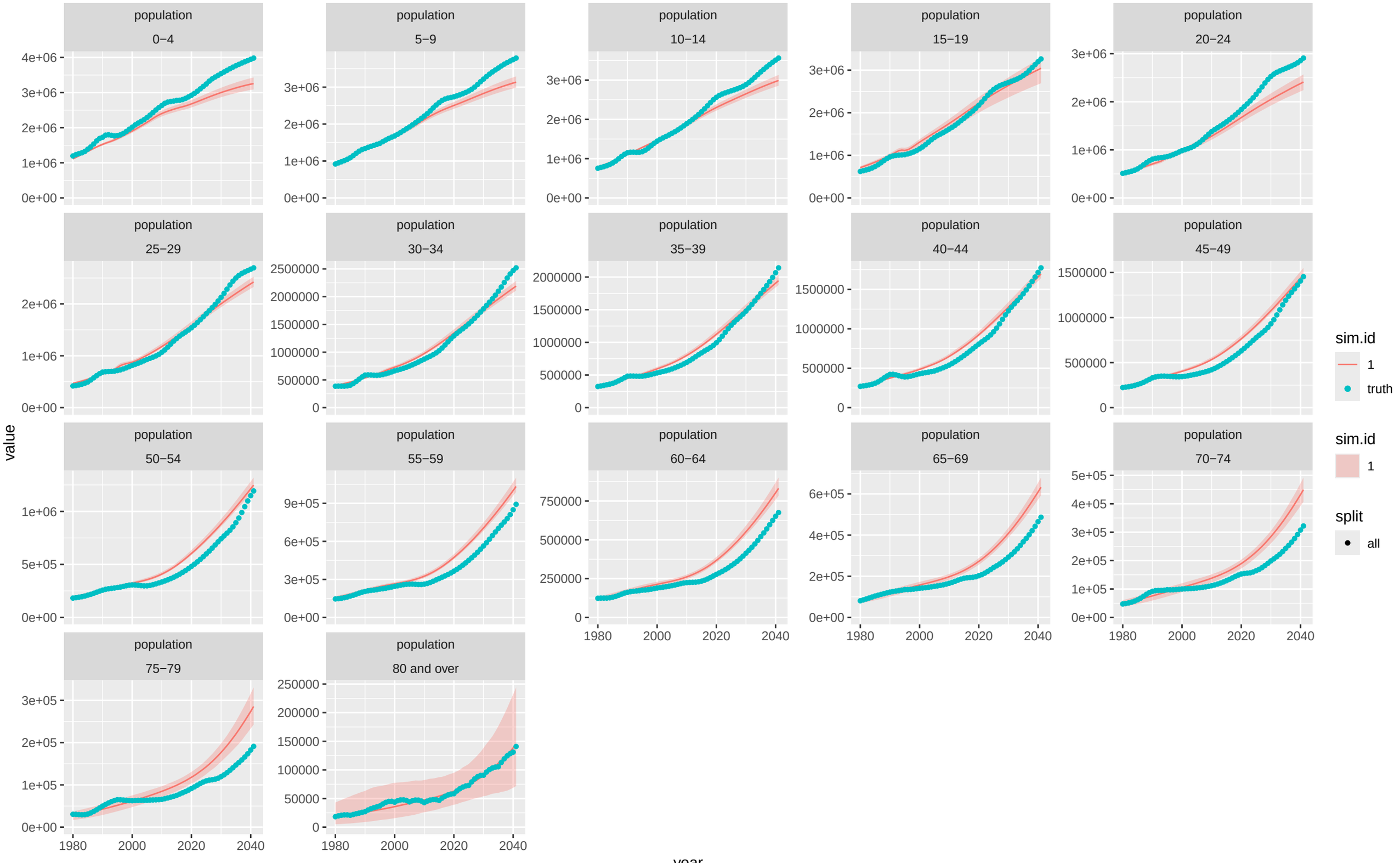
Malawi: incidence.by.age (merged Step 2 simset)



Malawi: prevalence.by.age (merged Step 2 simset)



Malawi: population.by.age (merged Step 2 simset)



Malawi: cascade (merged Step 2 simset)

awareness

engagement

suppression

value

1.00

0.75

0.50

0.25

0.00

1980

2000

2020

2040

year

sim.id

1

truth

sim.id

1

split

all

Malawi: cascade.by.age.sex (merged Step 2 simset)

